



3.3 Decision-Making Techniques

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Introduction to quantitative sales forecasting

- The sales forecast is an important **business planning tool**
 - It provides an **estimation of future sales figures** using **past data** and considering predictable **external factors**
- Sales forecasts can be used to **identify trends in product sales**, which can then be compared with the market as a whole

Methods used in quantitative sales forecasting

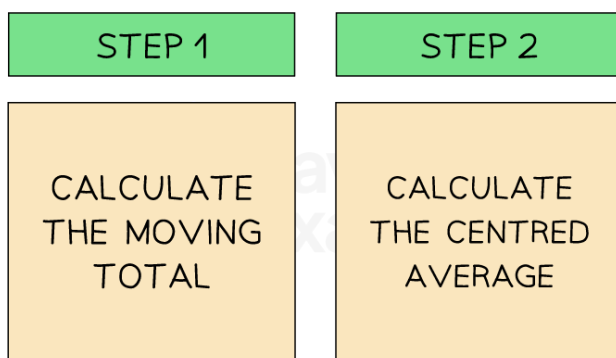
Method	Explanation
Moving averages	▪ A series of averages calculated from successive segments of a series of data ▪ These averages smooth data so that trends may be more easily identified
Extrapolation	▪ The prediction of future sales from past data ▪ Extrapolation can often be done simply by extending a line of best fit
Correlation	▪ Where there is a link between two variables, there is a correlation ▪ Correlations may be positive or negative

Calculation of time: Series analysis

Moving averages

- Sometimes, **past sales data is too erratic** for clear trends to be identified
- A moving average **smooths raw data** and allows analysts to **spot patterns**, even when sales are subject to **seasonal variations**
- Four-month or twelve-month moving averages are used where **seasonality** is a key factor in sales

Calculating moving averages



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Steps to calculate the moving average

- The moving total is calculated by **adding together sales figures for a specified number of periods**
 - E.g. a **three-month moving total** is calculated by **adding the first three months, followed** by the second three months and the third three months and so on
- The centred average is calculated by **dividing the moving total by the specified number of periods**
 - E.g. a **three-month centred average** is calculated by **dividing the three-month moving total by three**
- A **series of centred averages** is known as the **moving average**



Worked Example

RJ Inflatables is a manufacturer of novelty celebration balloons. Its monthly sales from January to July are shown in Table A.

Rachel Jameson, the managing director, is concerned that sales are declining but is struggling to identify a trend with the sales data she has available. Rachel's financial administrator has suggested using a moving average so that she can forecast future sales with greater accuracy.

Table A



Your notes

Month	Sales £
January	42,000
February	51,000
March	45,000
April	33,000
May	47,000
June	30,000
July	45,000

Calculate a three-month moving average using RJ Inflatables's January to July sales data.

(6)

Step 1: Calculate the three-month moving total of sales for each group of three months

Month	Sales (£)	Three-month Moving Total	
January	42 000	–	
February	51 000	–	
March	45 000	138 000	JANUARY + FEBRUARY + MARCH
April	33 000	129 000	FEBRUARY + MARCH + APRIL
May	47 000	125 000	MARCH + APRIL + MAY
June	30 000	110 000	APRIL + MAY + JUNE
July	45 000	122 000	MAY + JUNE + JULY

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Step 2: Calculate the three-month centred average for each group of three months

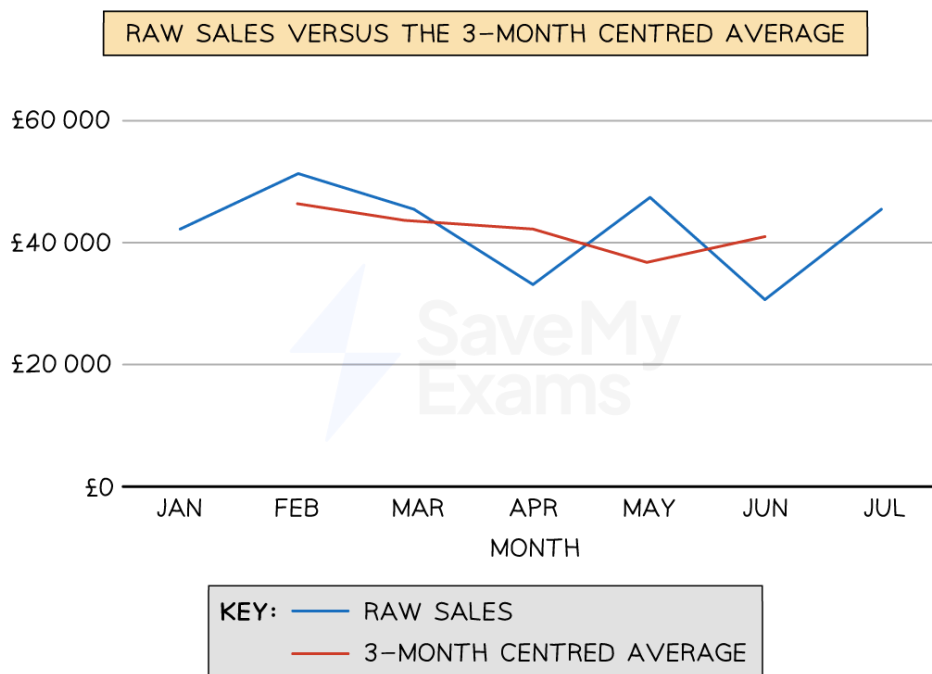


Your notes

Month	Sales (£)	Three-month moving total	Three-month centred-average	
January	42,000			
February	51,000		46,000	138,000 / 3
March	45,000	138,000	43,000	129,000 / 3
April	33,000	129,000	41,667	125,000 / 3
May	47,000	125,000	36,667	110,000 / 3
June	30,000	110,000	40,667	122,000 / 3
July	45,000	122,000		

- When plotted on the same graph, the three-month centred average provides a **smoother curve**, which makes **extrapolation** of the data for **forecasting** relatively straightforward

Raw data compared to the moving average



A comparison of raw sales data and 3-month centred average data

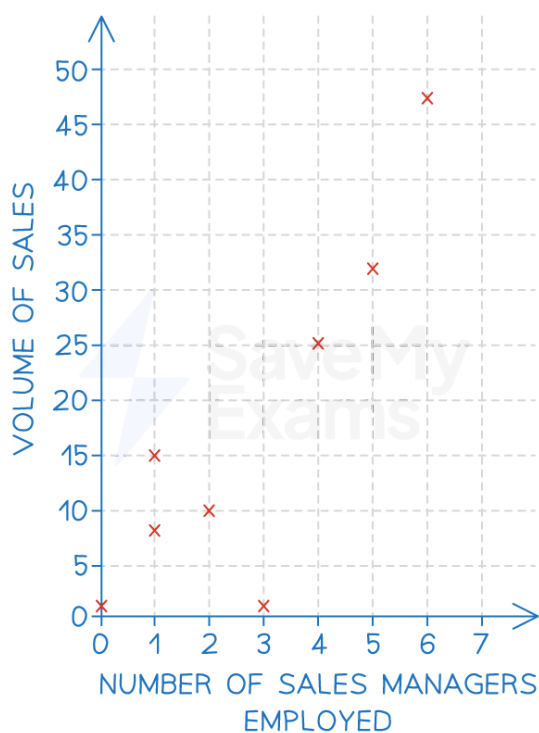
Interpreting scatter graphs

- Scatter graphs allow businesses to compare **two variables**, such as sales volume and advertising, to establish if there is any **correlation** between them

Example scatter graph



Your notes



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An example of a scatter graph showing the number of sales managers employed by a business and the volume of items sold

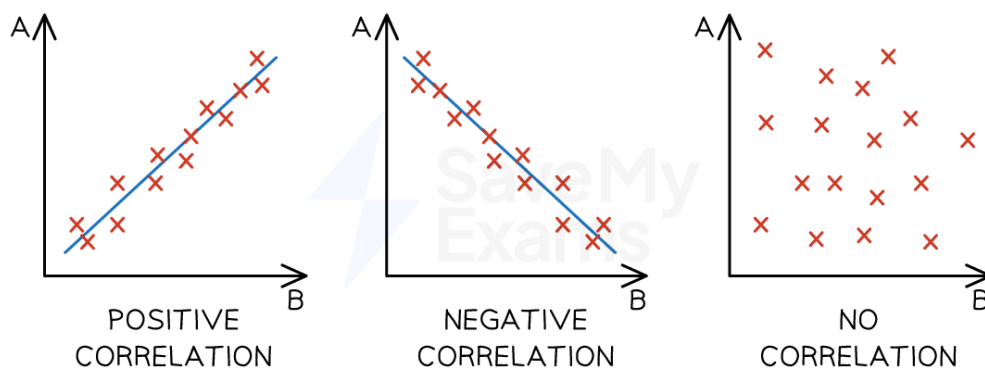
Types of correlation

- A correlation exists where there is a **relationship** or **connection** between two variables
 - A **positive correlation** means that **as one variable increases, so does the other variable**
 - A **line of best fit** that **slopes upwards** can be identified
 - A **negative correlation** means that **as one variable increases, the other variable decreases**
 - A **line of best fit** that **slopes downwards** can be identified
 - **No correlation** means that there is **no connection** between the two variables
 - Therefore, it is not possible to identify a **line of best fit**

Diagrams of correlation types



Your notes

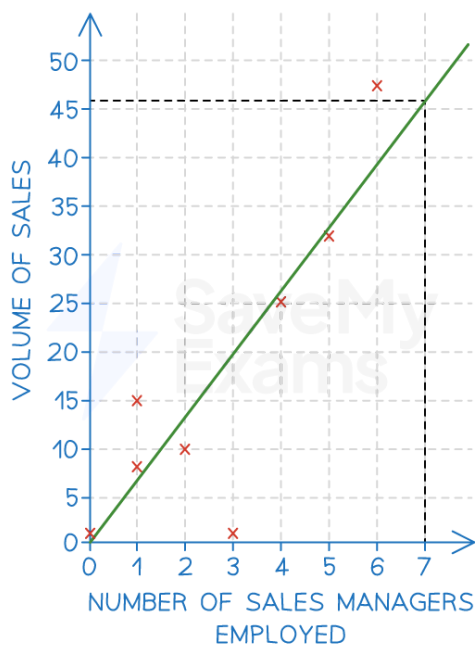


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The main types of correlation between two variables: positive, negative and no correlation

- Correlation does not always indicate a relationship or **causation** between two sets of variables, so businesses need to **conduct research** to establish **whether a relationship exists**, as well as the **strength of that relationship**
- Where a **line of best fit can be identified**, and when **causation** is determined, a business can **extrapolate** the data to make **predictions** around changes to either of the variables
 - E.g. **extrapolating** the line of best fit in the example below, the business could predict that **employing seven sales managers would likely result in sales of 46 units**
- Extrapolation assumes that what has **happened in the past** will be the same as what will happen in the future

Extrapolation using a line of best fit



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An example of a scatter graph with a line of best fit showing the number of sales managers employed by a business and the volume of items sold



Examiner Tips and Tricks

When drawing a line of best fit, you should try to include as many data points above the line as below the line.

Watch out for outlying data — if there is more than one outlier above the line, adjust your line of best fit upwards. Similarly, if there is more than one outlier below the line, adjust your line of best fit downwards. Just one outlier should not influence your line of best fit.



Your notes

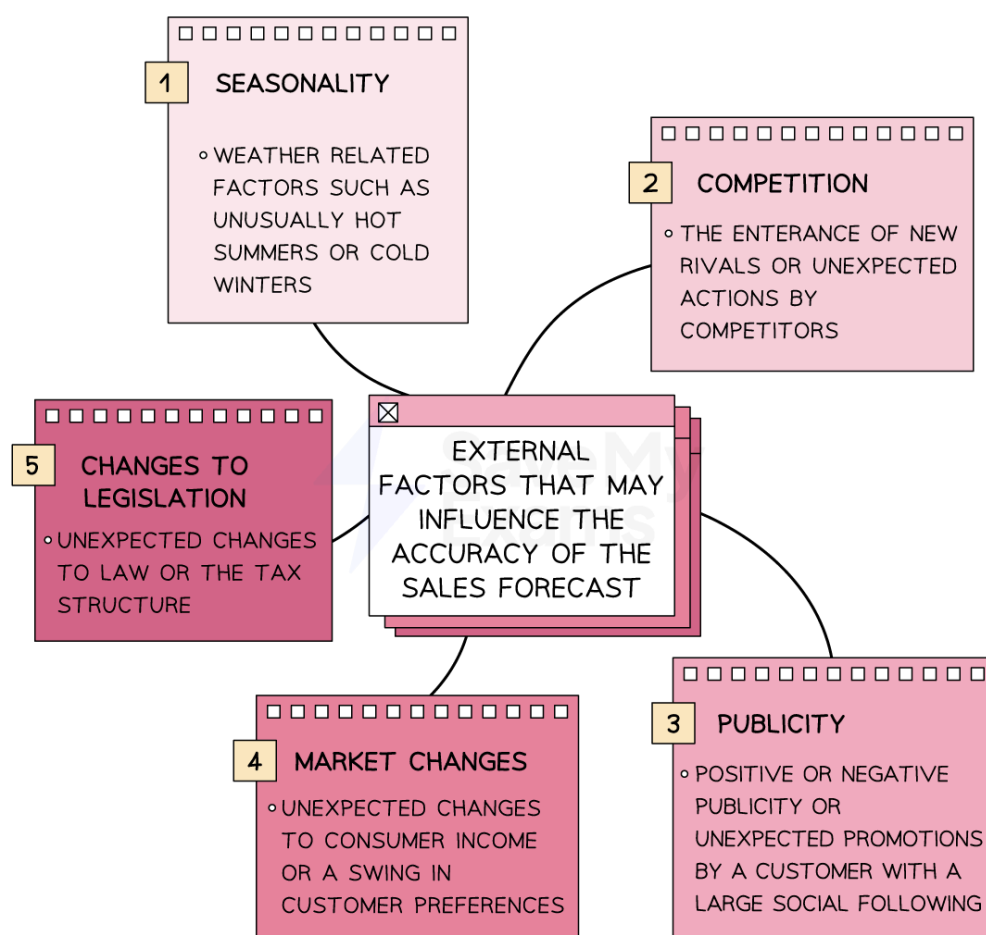
Limitations of quantitative sales forecasting

- Quantitative sales forecasting techniques are useful when the future is expected to reflect what has happened in the past
- It is especially effective in the **short term**
- In the longer term, uncertainties — especially in the **external environment** — can make simple extrapolation of past data unreliable

Factors affecting the accuracy of sales forecasts



Your notes



Examples of external factors that may influence the accuracy of the sales forecast include changes to legislation and market changes

- In many cases, the **sales forecast** can provide little more than an estimate of future performance
 - As long as it is **approximately accurate**, businesses can **use the sales forecast to plan resources**, such as staff, finance and production, and to **produce budgets**
- Businesses can improve the accuracy of sales forecasts by:
 - Conducting detailed **market research**
 - Employing experts with excellent **market knowledge**
 - **Revising** the sales forecasts frequently
 - Forecasting for the **short- to medium-term**



An introduction to investment appraisal

- Investment appraisal involves comparing **the expected future cash flows of an investment with the initial outlay** for that investment
- A business may want to analyse:
 - How soon** the investment will recoup the initial outlay
 - How profitable** the investment will be
- Before an investment can be appraised, key data will need to be collected, including:
 - Sales forecasts
 - Fixed and variable costs data
 - Pricing information
 - Borrowing costs
- The collection and analysis of this data is likely to **take some time**
 - It requires **significant experience** to interpret the data appropriately **before the investment appraisal can take place**
- Different methods are **used to appraise the value of an investment**, including:
 - The simple payback period
 - The average rate of return (ARR)
 - The net present value of the discounted cash flow

Simple payback period

- The payback period is a calculation of the **amount of time** it is expected an investment will **take to pay for itself**
- Where net cash flows are expected to be **constant over time**, the payback period can be calculated using the formula

$$\text{Payback period} = \frac{\text{Initial outlay}}{\text{Net cash flow per period}} = \text{Years / Months}$$



Worked Example

Gomez Carpets is considering an investment in a new storage facility at a cost of £200,000. It expects an additional net cash flow of £30,000 per year as a result of the investment.

Calculate the payback period for the investment.



Step 1: Divide the initial outlay by the additional expected net cash flow

$$= \frac{£ 200,000}{£ 30,000} \quad [1]$$

$$= 6.67 \text{ years}$$

Step 2: Convert the outcome to years and months

- 6 years
- 0.67 years = 8.04 months [1]
- Payback period = 6 years and 8 months [1]



Worked Example

Hammer and Son provides a household repairs service that has recently employed a new handywoman who requires her own van. The new van will be purchased for £32,000.

The net cash flows are expected to vary over the five years following its purchase and are shown in the table below.

Year	Net cash flow (£)	Cumulative cash flow (£)
0	(32,000)	(32,000)
1	14,000	(18,000)
2	10,000	(8,000)
3	6,000	(2,000)
4	3,000	1,000
5	2,000	3,000

Calculate the payback period for the van.

[4]

Step 1: Identify the final year where the cumulative cash flow is negative

- In this case, the cumulative cash flow figure is -£2,000 at the end of Year 3
- This is the remaining amount (outlay) outstanding [1]

Step 2: Calculate the monthly net cash flow for the next year

$$= £ 3,000 \div 12 \text{ (months)} \quad [1]$$

$$= £ 250$$

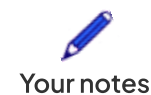
Step 3: Divide the remaining outlay outstanding by the monthly net cash flow

$$= \frac{£2,000}{£250} \quad [1]$$

= 8 months

Step 4: Identify the payback period

- In this case, the payback period is 3 years and 8 months [1]



Evaluation of the payback period method

Benefits	Drawbacks
- It is a simple method to calculate and understand - It is particularly useful for businesses where cash flow management is vital - Businesses can identify the point at which an investment is paid back and is contributing positively to cash flow - It is also useful when new technology is introduced regularly - Businesses purchasing equipment can calculate whether an investment "pays back" before an upgrade is available	- It provides no insight into the profitability of investments - Payback only considers the total length of time to recover an investment - Neither the timing nor the future value of cash inflows is considered - It may encourage a short-termism approach - Potentially lucrative investments may be dismissed, as they take longer to pay back than alternatives

Average rate of return

- The average rate of return (ARR) compares the average **profit** per year generated by an investment with the value of the **initial outlay**
- The average rate of return is calculated using the formula

$$ARR = \frac{\text{Average annual return}}{\text{Initial outlay}} \times 100$$

- The outcome of the formula is **expressed as a percentage**, which makes it easy to **compare different investment options**



Worked Example

Creative Frames, a small artwork framing business, is considering an investment of £40,000 in new machinery. Megan, the business owner, believes that total cash inflows over a six-year period will be £140,000 and total cash outflows will be £92,000.

Calculate the average rate of return of the proposed investment. [4]

Step 1: Calculate the total profit over the lifetime of the investment

$$\begin{aligned}\text{Total cash inflows} - \text{Total cash outflows} &= \text{Total profit} \\ &= £140,000 - £92,000 = £48,000\end{aligned}$$

Step 2: Divide the total profit by the number of years of the investment project to find the average annual profit

$$\begin{aligned}&= £48,000 \div 6 \text{ years} \\ &= £8,000\end{aligned}$$

Step 3: Divide the average annual profit by the initial outlay

$$\begin{aligned}&= £8,000 \div £40,000 \\ &= 0.2\end{aligned}$$

Step 4: Multiply the outcome by 100 to find the percentage

$$\begin{aligned}&= 0.2 \times 100 \\ &= 20\%\end{aligned}$$



Your notes

Evaluation of the average rate of return

Benefits	Drawbacks
- It considers all of the net cash flows generated by an investment over time - It is easy to understand and compare the percentage returns with each other	- As it depends on an average of cash flows, it ignores the timing of those cash flows - The opportunity cost of the investment is ignored, as values are neither expressed in real terms nor are adjustments made for the impact of interest rates and time

Net present value of discounted cash flow

- Net present value (NPV) is a financial metric used to evaluate the value of an investment or a project
- The NPV of an investment takes into account the **effects of interest rates and time**
 - It represents the **present value** of the **future cash inflows** minus the **present value** of the **future cash outflows**
 - To get the present value, the future value has to be discounted (reduced)
- This discounting method takes into account:



- The fact that money received in the future is worth less than money received today due to inflation
- The **opportunity cost** of not having the money available for other uses
- To calculate the NPV of an investment, **the value of all future net cash flows in today's terms** needs to be calculated first — and then **discounted using a table**
- The **cost of the initial investment is deducted** from the total of the discounted net cash flows:
 - If the sum of future net cash flows minus the initial investment is **positive**, then the investment is **likely to be worthwhile**
 - If the sum of future net cash flows minus the initial investment is **negative**, then the investment is **unlikely to be worthwhile**
- Discounted cash flows are calculated using **discount tables**, which allow future cash flows to be **expressed in today's terms**



Worked Example

Brownsea Sightseeing Tours Ltd is considering purchasing a new pleasure craft at a cost of £325,000.

It expects the investment to achieve the following net cash flows over five years of operation:

Year	Net cash flow (£)	10% discount factor
0	(325,000)	1.00
1	110,000	0.91
2	90,000	0.83
3	75,000	0.75
4	65,000	0.68
5	60,000	0.62

Using the 10% discount factor, calculate the NPV of the leisure craft investment.

Step 1: Calculate the discounted cash flow for each year by multiplying the net cash flow by the discount factor



Your notes

Year	Net cash Flow	10% Discount Factor	Discounted cash flow
0	(£325 000)	1.00	(£325 000)
1	£110 000	0.91	£100 100
2	£90 000	0.83	£74 700
3	£75 000	0.75	£56 250
4	£65 000	0.68	£44 200
5	£60 000	0.62	£37 200

$$£110\,000 \times 0.91$$

$$£90\,000 \times 0.83$$

$$£75\,000 \times 0.75$$

$$£65\,000 \times 0.68$$

$$£60\,000 \times 0.62$$

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Step 2: Total the discounted cash flow values for each year, including Year 0

$$= £325,000 + £100,100 + £74,400 + £56,250 + £44,200 + £37,200$$

$$= (12,550)$$

- The net present value of the investment is -£12,550
- Given that it is negative, this suggests that the investment in the new pleasure craft is **not financially worthwhile**

Evaluating the net present value method

Benefits	Drawbacks
▪ It considers the opportunity cost of money ▪ Discount tables are used to calculate the forecasted future values of net cash flows ▪ Businesses may choose different discount tables (20%, 10%, 5%, etc.) to adjust the level of risk involved in a project, allowing a range of scenarios to be considered	▪ It is more complicated to calculate and interpret than other methods of investment appraisal ▪ One of the primary challenges of using the NPV method is accurately forecasting future cash flows ▪ Selecting an appropriate discount rate can be challenging, and even small changes in the discount rate can significantly impact the calculated NPV ▪ The NPV method only considers the financial costs and benefits of a project and does not account for non-financial benefits or costs, e.g. environmental damage





Examiner Tips and Tricks

A common error made by students is the assumption that discounted cash flow takes into account the effects of inflation on investment decisions.

To account for inflation, the discount rate used in the NPV calculation should be adjusted to reflect the expected inflation rate.



Your notes

Limitations of investment appraisal techniques

- Each of the investment appraisal techniques **relies upon forecasted future cash flows**, which may **lack accuracy**:
 - Managers compiling cash flow forecasts may lack experience or may be biased towards a particular investment
 - Incomplete past data may make forecasting imprecise or mean that confidence in the data used to compile the forecast is limited
- **Long-term cash flow forecasts** can be **inaccurate** for several reasons:
 - Unexpected increases in **costs**
 - The arrival of **new competitors**
 - Changes in **consumer tastes**
 - Uncertainties due to economic growth or **recession**
- Factors other than the cost of investment and the return on investment are not considered:
 - Business finances and the availability of external finance to fund the investment
 - The overall **corporate objectives**
 - Potential for positive **public relations** or meeting **social responsibilities**



Decision tree diagrams

- A decision tree is a **quantitative method** of tracing the outcomes of a decision so that the most profitable decision can be identified
 - **Research-based estimates** and **probabilities** are used to calculate likely outcomes
 - The **net gain from a decision** can be identified and used to consider whether an investment is worthwhile
- Using decision trees provides **several key advantages** to the decision-making process:
 - Constructing a decision tree diagram may **reveal options** that haven't previously been considered
 - Managers are forced to **consider the risks** associated with their choice, ahead of implementation
 - The quantitative approach requires **deep research** to be carried out

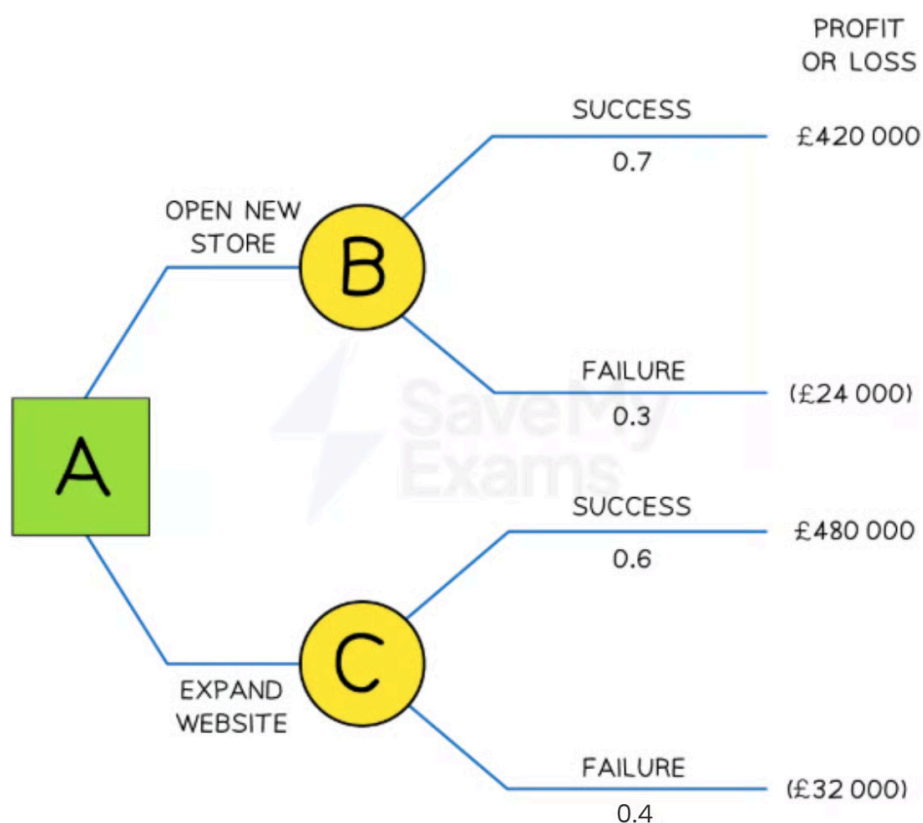
Decision tree diagrams

- The **key elements** in a decision tree diagram are:
 - Decision points
 - Outcomes
 - Probabilities
 - Expected monetary values

A simple decision tree



Your notes



A simple decision tree based on a choice of whether to invest in opening a new store or expand its website

- Points where **decisions** need to be made are represented by squares
 - Square A represents the fact that a choice is required on opening a new store or expanding the website
- Points where there are different **outcomes** are represented by circles called **nodes**
 - Circles B and C represent points at which the different options have a range of outcomes — success or failure
- The **probability** or likelihood of each outcome is shown on the diagram
 - A certain outcome has a probability of 1
 - An impossible outcome has a probability of 0
 - Opening a new store has a 0.7 probability of success and a 0.3 probability of failure
 - Expanding the website has a 0.6 probability of success and a 0.4 probability of failure
- The **monetary value** of each decision is based on the expected profit or loss of the outcome
 - If opening a new store is successful, a £420,000 profit is expected
 - If opening a new store is unsuccessful, a £24,000 loss is expected
 - If expanding the website is successful, a £480,000 profit is expected

- If expanding the website is unsuccessful, a £32,000 loss is expected

Calculating expected monetary values



Your notes

- To compare the options, a business should take into account the **expected values of each decision** presented in the decision tree diagram
- To calculate the **expected monetary value** of a decision, the following formula is used

(Expected value of success × probability) + (Expected value of failure × probability)

- Using the example above, the expected value of opening a new store is

$$\begin{aligned} & (£420,000 \times 0.7) + (-£24,000 \times 0.3) \\ &= £294,000 + -£7,200 \\ &= £286,800 \end{aligned}$$

- Using the example above, the expected value of expanding the website is

$$\begin{aligned} & (£480,000 \times 0.6) + (-£32,000 \times 0.4) \\ &= £288,000 + -£12,800 \\ &= £275,200 \end{aligned}$$

- As the expected value of opening a new store is higher, at £286,800, than that of expanding the website, at £275,200, then, based purely on financial terms, the business should choose the option to open a new store



Worked Example

Caramelac is a lactose-free chocolate product manufactured by a large multinational confectionery business. In recent years, increased competition from other well-known brands has started to impact sales of the product, and managers are determined to maintain Caramelac's market share.

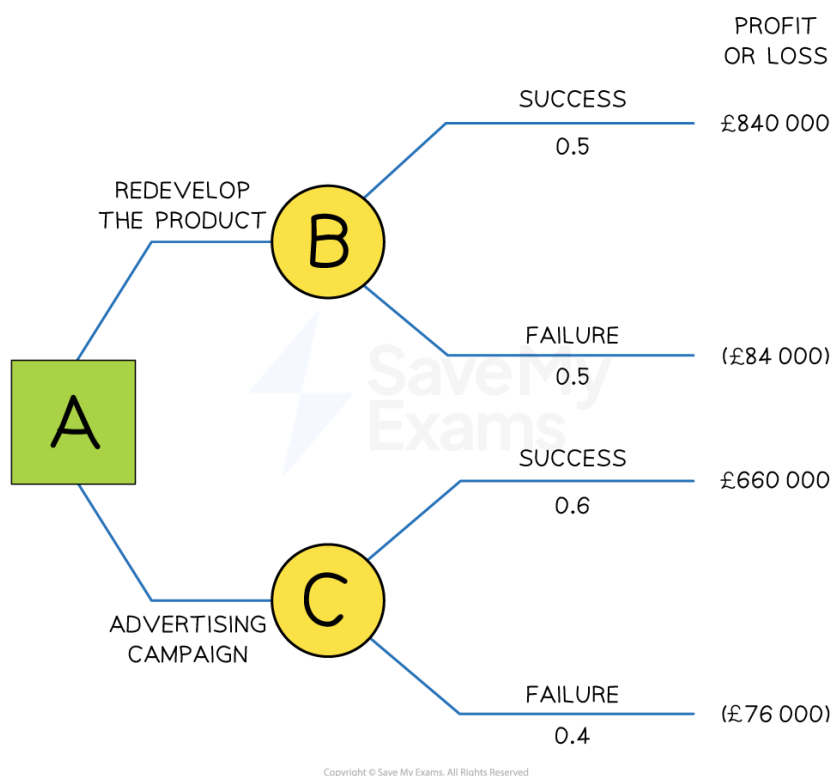
Market research has shown that the business has two options:

- Redevelop the product
- Create a new advertising campaign

The expected outcomes and the probabilities of success and failure are shown in the decision tree below.



Your notes



Calculate the expected values of each option and decide, on financial grounds, which option Caramelac's brand managers should choose.

[6]

Step 1: Calculate the expected value of redeveloping the product

$$= (£840,000 \times 0.5) + (-£84,000 \times 0.5) = £420,000$$

$$= £420,000 + -£42,000 \quad [1]$$

$$= £378,000$$

Step 2: Calculate the expected value of the advertising campaign

$$= (£660,000 \times 0.6) + (-£76,000 \times 0.4) = £396,000$$

$$= £396,000 + -£30,400 \quad [1]$$

$$= £365,600$$

Step 3: Interpret the outcomes and make a decision

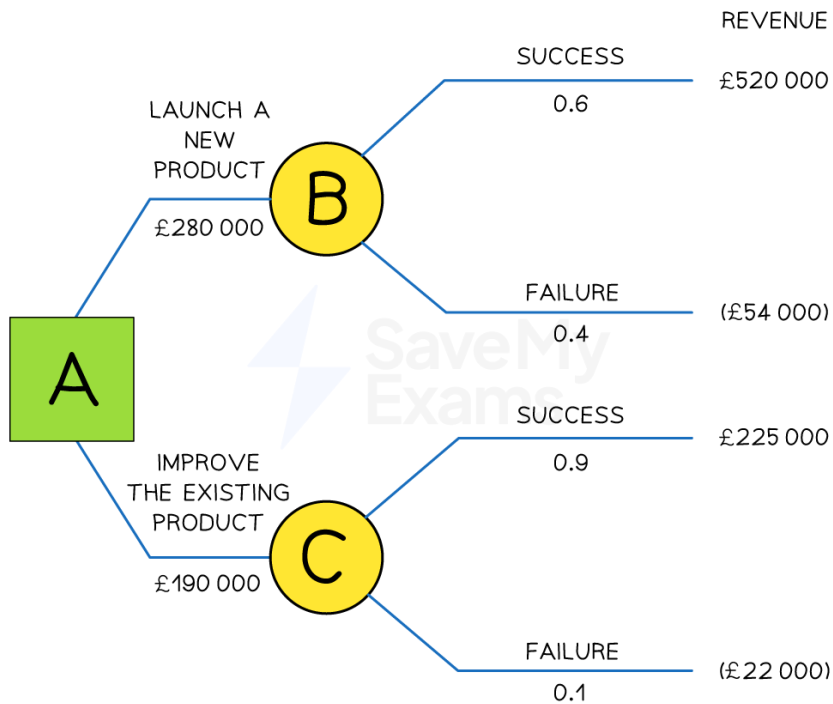
- As the expected value of redeveloping the product is higher, at £378,000, than that of the advertising campaign, at £365,600 [1], the business should choose the option to redevelop the product [1].



Your notes

Worked Example

- In some cases, the decision tree diagram **provides expected revenues rather than profit or loss** for the range of outcomes
- In these diagrams, **the costs related to each outcome are also provided**
- To calculate the expected value of each outcome, **costs must be deducted from expected revenues**



- To calculate the **expected monetary value** of a decision where revenues and costs are included in the diagram:

(Expected value of success $\times$ probability) + (Expected value of failure $\times$ probability) - cost

- The expected value of launching a new product is

$$= (£520,000 \times 0.6) + (-£54,000 \times 0.4) - £280,000 = £312,000$$

$$= £312,000 + -£21,600 - £280,000 = £290,400$$

$$= £290,400 - £280,000 = £10,400$$

- The expected value of improving the existing product is

$$= (£225,000 \times 0.9) + (-£22,000 \times 0.1) - £190,000 = £202,500$$

$$= £202,500 + -£2,200 - £190,000 = £200,300$$

$$= £200,300 - £190,000 = £10,300$$



Your notes

- As the expected value of launching a new product is marginally higher, at £10,400, than that of improving the existing product, at £10,300, the business should choose the option to launch a new product
- In this case, the decision tree has demonstrated that there is **little difference between the two options**, and the business should **look at other factors** that may inform its decision



Examiner Tips and Tricks

Expected values are not the same thing as profit or revenues generated by one of the choices. In the above example, launching a new product is expected to either generate a positive revenue figure of £520,000 or generate a negative revenue figure of £54,000. It is never forecast that a revenue figure of £200,300 will be achieved. This is purely a figure used in making the choice between this option and the alternative and does not represent the actual amount of revenue that is expected to be achieved.

Limitations of using decision trees

- Constructing decision trees that can support effective decision-making **requires skill** to avoid bias and takes a **significant amount of time to gather reliable data**
- A decision tree is **constructed using estimates** that rarely take full account of **external factors** and **cannot include all possible eventualities**
- **Qualitative elements**, such as human resource impacts, are not considered, which may **affect the probability of success** of a decision
- The **time lag** between the construction of a decision tree diagram and the implementation of the decision is likely to further affect the **reliability of the expected values**



The nature and purpose of critical path analysis

- Critical path analysis is a **project management tool** that uses **network analysis** to plan **complex** and **time-sensitive** projects
- Critical path analysis involves the construction of a visual model of the project that includes key elements:
 - A list of all **activities** required to complete the project
 - The time (**duration**) that each activity will take to complete
 - How each project activity **depends** on others
- Critical path analysis shows:
 - The order in which activities must be completed
 - The **longest path of project activities** to the completion of the project
 - The **earliest** and **latest** that each project activity can **start and finish** without delaying completion of the project as a whole
 - Activities within a project that can be carried out simultaneously
 - The **critical project activities** that, if delayed, will cause the project as a whole to overrun
 - Those project **activities where some delay is acceptable** without delaying the project as a whole
 - The **shortest time possible** to complete the project
- It allows managers to identify the **relationships between the activities** involved and to work out the **most efficient way of completing the project**
 - **Resources** such as raw materials and components can be ordered or hired at precisely the right time they are needed
 - **Working capital** may be managed efficiently
 - Where **delays** occur, managers can **identify the implications for the project's completion** and **redirect resources** if required

Drawing critical path analysis diagrams

Components of network analysis diagrams

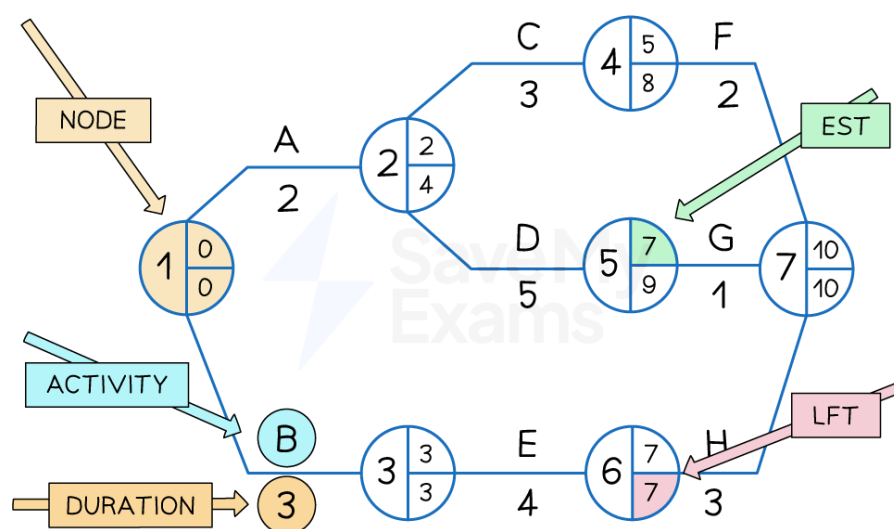
Element	Description
Node	▪ A node is a circle that represents a point in time where an activity is started or finished ▪ The node is split into three sections



Your notes

	▪ The left half of the circle is the activity number ▪ The top right section shows the earliest start time (EST) that an activity can begin based on the completion of the previous activity ▪ The bottom right section shows the latest finish time (LFT) by which the previous activity must be completed
Activities	▪ An activity is a process or task within a project that takes time ▪ Activities are shown on the network diagram as a line that links nodes ▪ A description of the activity or a letter representing the activity is usually shown above the line
Duration	▪ The duration is the length of time it takes to complete an activity ▪ The duration is shown as a number of time units, such as hours or days, below the activity line

Example network diagram



A simple network diagram showing key elements: node, activity and duration

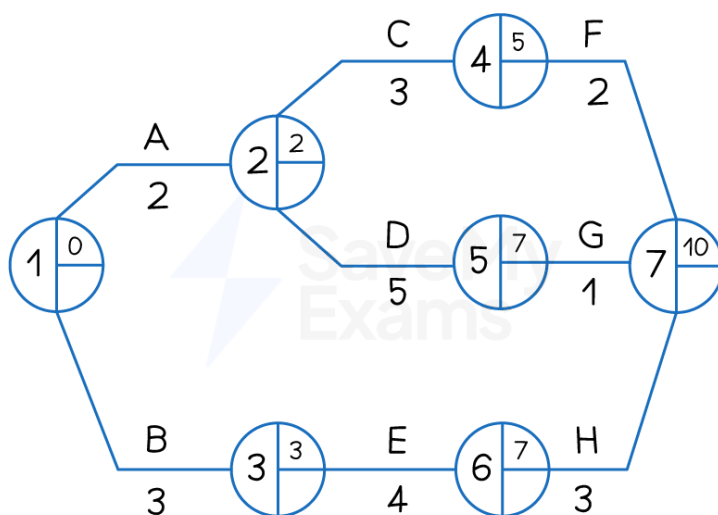
- A network diagram must **always** start and end at a single node
- Lines must not cross and must **only** be assigned to activities

Critical path calculations

Earliest start times



Your notes



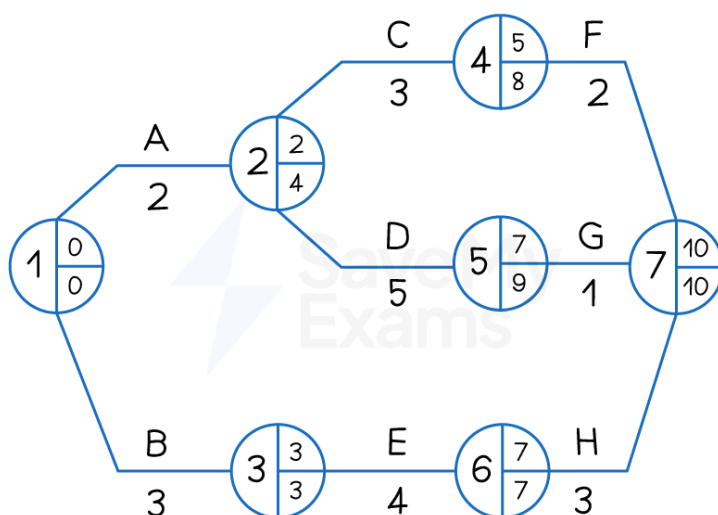
An example of a simple network diagram showing earliest start times

- Working forwards from Node 1, it is possible to calculate the **earliest start time (EST)** for each activity by **adding the duration of each task**
- The EST for each activity is placed in the **top right** of each node
 - Node 1 is the starting point of the project and where both Activity A and Activity B begin
 - Activity A and Activity B are independent processes
 - Activity A has a duration of two days, and its EST is zero days
 - Activity B has a duration of three days, and its EST is also zero days
 - Activity C and Activity D both begin at Node 2 and are dependent upon the completion of Activity A but are independent from each other
 - Activity C has a duration of three days, and its EST is two days
 - Activity D has a duration of five days, and its EST is also two days
 - Activity E begins at Node 3
 - Activity E has a duration of four days, and its EST is three days
 - Activity F begins at Node 4
 - Activity F has a duration of two days, and its EST is five days
 - Activity G begins at Node 5
 - Activity G has a duration of one day, and its EST is seven days
 - Activity H begins at Node 6
 - Activity H has a duration of three days, and its EST is seven days
 - Node 7 is the endpoint of the project

Latest finish times



Your notes



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An example of a simple network diagram showing ESTs and LFTs

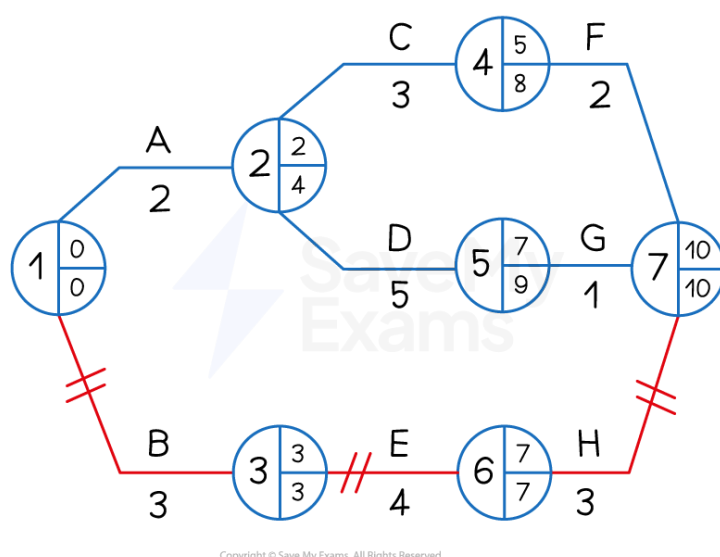
- Working backwards from Node 7, it is now possible to calculate the latest finish time (LFT) for each activity by **subtracting the duration of each task**
- The LFT for each activity is placed in the **bottom right** of each node
 - Node 7 is the endpoint of the project, which has an LFT of ten days
 - Activity H has a duration of three days
 - The LFT in Node 6 is seven days (10 days – 3 days)
 - Activity G has a duration of one day
 - The LFT in Node 5 is nine days (10 days – 1 day)
 - Activity F has a duration of two days
 - The LFT in Node 4 is eight days (10 days – 2 days)
 - Activity E has a duration of four days
 - The LFT in Node 3 is three days (7 days – 4 days)
 - Activity D has a duration of five days
 - The LFT in Node 2 is four days (9 days – 5 days)
 - Activity C has a duration of three days
 - The LFT in Node 3 is four days because Activity D is the more time-critical of the two activities that are dependent upon the completion of Activity A, so its LFT is recorded
 - Activity B has a duration of three days
 - The LFT in Node 1 is zero days (3 days – 3 days)
 - Activity A has a duration of two days



- The LFT in Node 1 is zero days because Activity B is the more time-critical of the two starting activities, so its LFT is recorded
- The LFT in Node 1 is always zero

Identifying the critical path

- The critical path highlights those **activities that determine the length of the whole project**
 - If any of these critical activities are delayed, the **project as a whole will be delayed**
- The critical path follows the **nodes where the EST and LFT are equal**
 - In the diagram below, Nodes 1, 3, 6 and 7 have equal ESTs and LFTs
 - Activities that determine these nodes are B, E and H
 - These activities are marked with two short lines
 - The critical path is therefore BEH



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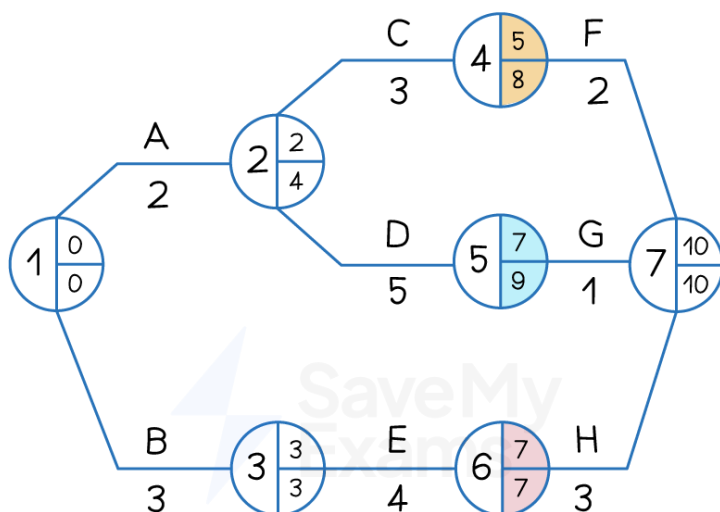
An example of a simple network diagram showing the critical path, BEH

Identifying and calculating float time

- Float time exists where there is a **difference between the EST and the LFT**
- Where float time is identified, managers may:
 - **Transfer resources**, such as staff or machinery, to more critical activities
 - **Allow extra time** to complete tasks to improve quality or allow for creativity



Your notes



AT NODE 4 THERE IS A DIFFERENCE OF 3 DAYS (8 DAYS – 5 DAYS)

AT NODE 5 THERE IS A DIFFERENCE OF 2 DAYS (9 DAYS – 7 DAYS)

AT NODE 6 THERE IS NO FLOAT TIME. SPARE RESOURCES MAY BE USED TO SUPPORT COMPLETION OF CRITICAL ACTIVITY H

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An example of a simple network diagram showing float nodes (4 and 5) and a critical node (6)

- The **total float** refers specifically to the spare time that is available so that the **overall project completion is not delayed**
- The **total float** for a specific activity is calculated using the formula

LFT for the activity – Duration of the activity – EST for the activity

- Using the diagram above, the following **total float times** can be calculated for Activities A to H

Activity	LFT	- Duration	- EST	= Total float
A	4	2	0	2
B	3	3	0	0
C	8	3	2	3
D	9	5	2	2
E	7	4	3	0
F	10	2	5	3

G	10	1	7	2
H	10	3	7	0



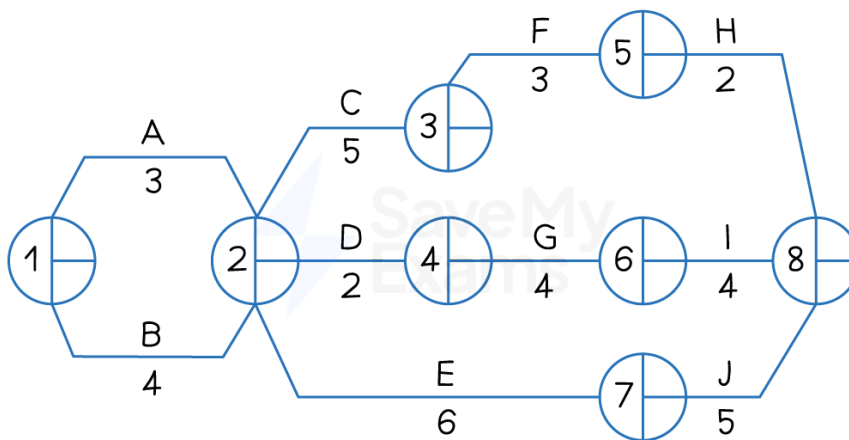
Your notes

- The critical activities B, E and H each have a total float of zero days



Worked Example

The network diagram below shows the activities involved in a new promotional campaign for a small fashion accessories business, as well as the time (in weeks) it is expected that each activity will take to complete.



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Calculate:

- The ESTs and LFTs for each node [4]
- The total float time for Activity G [3]

Step 1: Calculate the ESTs

Node 1 EST = 0

Node 2 EST = $0 + 3 = 3$, but $0 + 4 = 4$ so 4

Node 3 EST = $4 + 5 = 9$

Node 4 EST = $4 + 2 = 6$

Node 5 EST = $9 + 3 = 12$

Node 6 EST = $6 + 4 = 10$

Node 7 EST = $4 + 6 = 10$

Node 8 EST = $12 + 2 = 14$, but $10 + 4 = 14$, and $10 + 5 = 15$, so 15 [2]

Step 2: Calculate the LFTs



Your notes

Node 8 = 15

Node 7 = 15 - 5 = 10

Node 6 = 15 - 4 = 11

Node 5 = 15 - 2 = 13

Node 4 = 11 - 4 = 7

Node 3 = 13 - 3 = 10

Node 2 = 10 - 6 = 4

Node 1 = 4 - 4 = 0 [2]

Step 3: Calculate the total float time for Activity G

Total float = LFT for the activity - Duration of the activity - EST for the activity

= 11 weeks - 4 weeks - 6 weeks

= 1 week

[3]

Limitations of using critical path analysis

Limitations	Explanation
▪ Very lengthy or complex projects involve a very large number of activities that have numerous dependencies	▪ Specialist network planning software may be required ▪ Layers of supervision may be required to manage different groups of activities within the project
▪ Network analysis often relies on estimates and forecasts	▪ Significant research is required prior to the completion of network analysis ▪ Close and honest working relationships with suppliers are essential
▪ Network analysis does not guarantee the success of a project	▪ Project managers will need to be highly skilled and will need experience working with complicated plans
▪ Resources may not prove to be as flexible as hoped when managers identify float periods	▪ Employees may require additional training in order to transfer to critical tasks ▪ Machinery and other capital resources may require adaptation